

SUSHANT LOKHANDE

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EXPERIENCE

San Jose State University | Graduate Research Assistant, Retrieval & ML Aug 2025 to May 2026

- Built and evaluated **RAG** pipelines (chunking, embeddings, vector retrieval, top-k, grounded prompts), separating **retrieval** failures from generation failures to guide fixes.
- Ran **evals** on RAG workflows by comparing chunking strategies, embedding models, and top-k settings, and classified each failure as retrieval-caused or generation-caused.
- Boosted malware classification to **80.2%** accuracy across 17 families with a soft-voting **PyTorch** ensemble (CNN, ResNet50, ViT), **2.4 pts** over the best single model at $p < 0.001$.
- Cut experiment setup effort by about **55%** by automating orchestration, configuration, **telemetry** collection, and analysis, making large experiment sweeps reproducible.

Turito | Software Engineering Intern, Backend & Real-Time Jun 2022 to Oct 2022

- Cut controlled-disruption failures by **42%** (3.1% to 1.8%) by hardening **retries**, timeouts, failure recovery, and distributed state transitions.
- Slashed p95 **CPU** by **22%** by profiling high-frequency media paths and removing excess buffer allocation, payload copies, repeated routing lookups, and **lock contention**.
- Reduced production diagnosis time by **35%** by adding structured **metrics**, runtime-state visibility, and targeted logs.
- Worked across a **WebRTC** control plane and **RTP/RTCP** media paths spanning about **12 regions**, including ICE/STUN/TURN connection setup and degraded-network handling.

Trivia Softwares | Software Engineering Intern Nov 2021 to Jan 2022

- Built a **Rust/Tokio** load-testing framework driving about **25,000** concurrent sessions, exposing lock contention, serialized **WebSocket** writes, and slow session cleanup.
- Reduced performance-investigation time by about **40%** with tooling built on Rust workers, **Python** aggregation, **SQLite**, Flask, and Chart.js.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, SQL

LLM & AI: RAG, embeddings, evals, agents, tool use, LangChain, OpenAI and Anthropic APIs, PyTorch, Hugging Face

Backend & Frontend: FastAPI, asyncio, SSE, REST, gRPC, React, Next.js, Node.js

Data & Infra: PostgreSQL, Redis, Qdrant, SQLite, Kafka, Temporal, Docker, AWS, pytest

PROJECTS

Relay, LLM Inference Gateway | *Python, FastAPI, asyncio, SQLite, SSE*

- Built an OpenAI-compatible gateway with exact and **semantic caching**, provider **failover**, and rate limiting that cut p50 latency from **759 ms to 44 ms** over a 4,000-request load test.
- Served **78%** of requests from cache and cut modeled provider cost by **76%**, with live p50/p95, token, and cost metrics.

Teller, Agent Evaluation Environment | *Browser agents, Python, TypeScript*

- Built an adversarial target app (nested frames, missing test IDs, session expiry, permission denials) and analyzed **14 agent runs**, classifying failures as planning, execution, validation, or recovery.

Proxima, Vector Search Engine | *C++17, AVX2/SIMD, HNSW, pybind11*

- Built an HNSW vector search engine from scratch that reaches **0.999 recall** at **12,049 QPS** single-threaded, **1.8x** faster than hnswlib and **2.5x** faster than FAISS on 100K vectors.

OPEN SOURCE

- NVIDIA/garak**: preserved attempt UUIDs in report aggregation for traceable LLM evals (merged, shipped in v0.17.0)
- microsoft/markitdown**: fixed EPUB conversion for percent-encoded manifest paths, with regression tests (merged)

EDUCATION

San Jose State University | M.S. in Computer Science | GPA 3.8/4.0 | Spartan Scholar Aug 2024 to May 2026

Coursework: Machine Learning, Artificial Intelligence, Distributed Computing, Database Systems, Advanced Algorithms, NoSQL

University of Mumbai | B.Tech. in Computer Engineering | GPA 3.9/4.0 Jun 2019 to May 2023

Research: malware image-classification ensemble selected for Dr. Mark Stamp's ML textbook; sole-author contactless palmprint verification (0.9975 AUC)